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A novel knowledge-driven flexible human–robot hybrid disassembly line and its key technologies for electric vehicle batteries☆

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ABSTRACT

Based on the unique problems and challenges in the disassembly scenario of waste electric vehicle batteries (EVBs), we propose a knowledge-driven flexible human–robot hybrid disassembly line. The disassembly line can not only split the EVB disassembling tasks layer by layer to the primitive-level subtasks based on knowledge but also intelligently generate the optimal disassembling primitive sequence to complete the disassembling task. To further elaborate on the key technologies of the above-mentioned disassembly line, we focus on the implementation of the high-accuracy screw disassembly and disassembly planning system based on NeuroSymbolic. Implementation of the high-accuracy screw disassembly is realized by a new design of the disassembly actuator, high-accuracy screw position and pose estimation based on visual perception, and robust insertion based on force perception. The disassembly planning system is constructed by defining disassembly primitives and introducing neural predicates which can take the sensor data as input and output the symbolic states. Experiments show that our system can effectively solve the screw disassembling task in complex disassembly scenarios.

1. Introduction

Due to the vigorous development of the electric vehicle industry, electric vehicle batteries (EVBs) as large-scale electric storage devices, have been widely applied for energy supplement because of their outstanding characteristics of small size, high voltage, and energy density [1]. With increasing concerns regarding environmental issues and limited supplement of lithium and cobalt, recycling of EVBs is imperative for the healthy and sustainable development of the electric vehicle industry [2].

Currently, there are two mainstream methods for the recycling of EVBs, mechanical methods and metallurgy methods [3], in which the former refers to disassembling battery packs into modules, modules to cells as well as the subsequent crushing and sorting operations while the latter refers to using pyro-, hydro-, bio-metallurgy and hybrid methods. Given that conducting the disassembly operations preferentially brings many substantial benefits such as lower safety risks and higher purity of recycled materials, mechanical methods, especially the process of disassembling battery packs into modules, modules to cells, inevitably act as the preliminary and essential step for subsequent operations [4].

Nowadays, the huge uncertainties in types, design complexity and end-of-life(EOL) states [5] of the disassembly object have posed a huge challenge to the automatic disassembly of EVBs. The type and design complexity of the disassembly object and the disassembly goal corresponding to different EOL states put forward the requirement for the reasonable selection of disassembly tools and methods, and the dexterous operation like disassembling the screw fasteners and covers of bus bars (snap-fit structures) also creates difficulties for automation. Therefore, the current mechanical disassembly of EVBs is still dominated by manual disassembly at the cost of low efficiency and great harm to workers [6]. In order to meet the needs of safe, efficient and flexible disassembly, the disassembly methods of EVBs urgently need to be changed from manual disassembly to automatic and intelligent disassembly.

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In order to realize the intelligent disassembly of EVBs under an unstructured disassembly environment, we propose a knowledge-driven flexible human–robot hybrid disassembly line. The disassembly line can not only split the EVB disassembly task layer by layer to the primitive level based on knowledge but also intelligently generate the disassembling primitive sequence through NeuroSymbolic task and motion planning (NeuroSymbolic TAMP) system [7–9]. As a necessary module for direct control of robotic disassembly operations, the NeuroSymbolic TAMP system uses Planning Domain Define Language (PDDL) [10–12] to define each disassembly primitive of the entire procedure. In particular, we introduce the neural predicates [9,13] to describe continuous physical states, taking the sensor data as input and outputting the symbolic states. In this way, the whole system can adapt to the complex disassembling environment and make autonomous decisions within a symbolic logic framework. The NeuroSymbolic TAMP system knows what and why the robot is doing, which is very important in the context of human–robot collaboration.

To further elaborate on the implementation details of the key technologies of the above-mentioned disassembly line, we choose the task of disassembling screws of the EVBs as the main case for illustration. On the one hand, it is unscrewing operations that are considered as the most necessary and most likely steps to be automated [14,15], taking up approximately 40% of all disassembly activity [16,17]. Therefore, realizing the task of disassembling screws intelligently in an unstructured disassembling environment will greatly reduce the workload of workers and facilitate the disassembly process of EVBs, which is of great significance. On the other hand, combined with relevant literature research [5] on the technical readiness of each link in the EVB disassembly process, there is still a lot of room for improvement in the disassembly operation, dexterous robotic manipulations, and human–robot cooperation. More specifically, compared with the previous research [9] which focused on the feasibility analysis of NeuroSymbolic with verification experiments based on simulation, this paper focuses on complementing and optimizing the original content to ensure the successful disassembly operations of robots in actual scenarios. New content includes extensions to disassembly primitives, high-accuracy fastener position and pose estimation algorithms based on visual perception, robust insertion algorithms based on force perception. Our paper will provide new ideas for the corresponding technical links.

The remainder of this paper is structured as follows. Section 2 reviews the relevant studies about intelligent disassembly. Section 3 analyzes the disassembling process of EVBs and proposes a new framework of knowledge-driven flexible human–robot hybrid disassembly line. Relevant key technologies about screw disassembly would be elaborated in Section 4. Correspondingly, a series of experiments are conducted to verify the efficiency and reliability of our proposed methods in Section 5. Finally, the conclusion and future work are discussed in Section 6.

2. Literature review

This section reviews relevant literature which is divided into four key areas of relevance. The first two subsections review the existing automatic disassembly systems and the development of dexterous robotic manipulations for disassembly. Small fastener object detection is discussed in the third subsection. And the fourth subsection presents current work in the research of systems with autonomous adaptability to dynamic environments.

2.1. Automatic disassembly systems

Aiming at the automatic disassembly of EVBs, some scholars [17–19] have proposed to establish a human–robot disassembly workstation and assign highly repetitive tasks to robots according to the difficulty of disassembling tasks. Optisort system [20,21], an automatic sorting system for consumer batteries, utilizes machine vision algorithms to identify labels on batteries and sort batteries by their chemical composition. Automatic disassembly systems have also been implemented in other sectors. For example, Apple established an automatic disassembly line Liam [22], which can disassemble an iPhone 6 in 11 s, with an annual recycling volume of 1.2 million units.

Although the above-mentioned automatic disassembly systems replace some of the manual operations, these systems are far from reaching the goals of intelligent disassembly owing to the lack of flexibility and adaptability to deal with random disassembly uncertainties.

2.2. Dexterous robotic manipulation

From the perspective of dexterous manipulations, it is a worldwide challenge of robot research [23]. For intelligent disassembling systems, dexterous robotic manipulation is also an important part.

Recently, many subtle disassembly systems have been developed leveraging multiple sensors and new algorithms. A highly integrated and compact scheme [24,25], which is implemented by a single arm and exchangeable tools, is proposed for disassembling operations that generally require bimanual manipulation. Beyond designing entire disassembly systems, many sophisticated end effectors [26–30] are fabricated to achieve specific tasks and obtain accurate grasping pose through certain well-designed features and robust algorithms. Despite breakthroughs in mechanics with sensors for perceiving force and new mechanics for soft actuation to offer natural compliance, these systems are less capable of complying with uncertainties in the disassembly environment.

Robot learning is crucial for dexterous robotic manipulation in an unstructured and uncertain disassembly environment, which can typically be formulated as individual Markov Decision Processes(MDPs) [31–33]. Generally, MDPs refer to state, observation, action, transition function, a policy assigning an action to a transition, and cost or reward function for the policy [34]. Several studies [35–38] have investigated successful applications of MDPs to solve motion trajectory optimization and construct highly robust executable solutions. Taking a step further on the basis of MDPs, reinforcement learning(RL) [39] is widely used to learn policy parameters for skill controllers in robotics domains. However, there are many challenges to applying RL to robotics. Since many tasks in robotics are multi-objective (e.g.,pick up the mug and do not collide with anything), it can be difficult to define appropriate reward functions that obtain the desired behavior [40–42]. Due to the episodic nature of most robotics tasks, rewards tend to be sparse and difficult to learn from. Robotics problems also have high-dimensional continuous state features, as well as multi-dimensional continuous actions, which make policy learning challenging. Moreover, RL requires robots to learn corresponding skills through continuous attempts in the environment. However, the uncertainty of battery disassembly is high, and it is difficult to build a corresponding environment in advance for robots to learn.

Most notably, the new developments leverage the immense progress in machine learning to encapsulate models of uncertainty and support further advances in adaptive and robust control [23,43]. One grasping policy [44] trained on the Dex-Net 4.0 dataset with deep neural network(DNN) is developed for universal and dexterous robotic picks and grasps, reaching 95% reliability with 300 picks per hour. The robustness of some other dexterous manipulations like peg-in-hole insertion [45,46] is also enforced by the applications of DNN-based methods. To further elaborate on data-driven methods but avoid generating examples with real, physical systems, many researchers [44,46] use simulation environments and generate a large amount of synthetic data for data diversity.

Despite immense advances combining sophisticated hardware design and delicate control algorithms, robotic manipulations are still generally much below human handling dexterity [23]. More researches and applications need to be further investigated.

2.3. Small fastener object detection

The essence of object detection process is to learn the mapping between the sensing data and the state of the disassembly objects. As a result, prospective computer vision technologies are widely applied in the field of robots and automation. One preliminary attempt [18] is to utilize Haar Cascade classifier [47] from OpenCV library identifying the relevant bolts, but this approach is totally incapable of sorting the identified bolts according to their categories. Another intriguing implementation [48] designs a robotized workstation for automatic disassembly with a particular procedure for target detection based on the grayscale, depth and HSV values. More modal information is introduced to filter the false detection results in this way. Similarly, a series of subsequent thresholding and morphological filtering operations [26] provides a solution for detecting centers of screw holes. The above methods are generally conducted through extracting and classifying artificial features with limited performance of detection, classification and computing time.

Meanwhile, a new paradigm of machine vision centered on deep learning has shown powerful capabilities in performing the detection [49]. There are two generic frameworks for implementing detection based on Convolutional Neural Network(CNN). One is designed to first generate and select candidate regions in one image and then perform convolution and other processing operations in these regions, including Region-CNN(R-CNN) [50] and its variants [51–53]. The other one treats the detection of the image as a whole global regression in real-time. In particular, You Only Look Once(YOLO) [54] and its variants are the most representative frameworks. Some preliminary researches have demonstrated their great potential for real-time detection of tiny fasteners in disassembly scenarios, especially in the disassembly of EVBs. To distinguish the screws with similar geometric shapes, an accurate screw detection method [55], incorporating Faster R-CNN and an innovative rotation edge similarity algorithm, is proposed and achieves satisfactory classification accuracy of 99.64%, while the detection speed is relatively slow at 4.98s per screw. Furthermore, Faster-RCNN detection modules are applied as the core perception units of the information-driven robotic disassembly system [56], which demonstrate a detection accuracy of 73.71% and average localization precision of 0.86 mm for $TX30\_M6 \times 12$ screws. Compared with the R-CNN family, YOLO and its variants have high speed superiority for real-time detection. A recent study [57] utilized YOLO v3 for the detection of different components in an EVB, such as screws, battery modules, connecting plates, and so on.

Apparently, the above mentioned methods can be divided into two main paradigms, traditional methods and deep learning methods. Each of the methods has merits and shortcomings, the former is relatively independent of the relevant data and avoids the time-consuming collection and labeling of the data. Conversely, the latter requires a high quality data set to ensure the performance of algorithms and robustness under complex conditions. Considering the real-time requirements for detection in industrial disassembly scenarios, more applications of deep learning methods are reasonably likely to be foreseen in the future.

2.4. Intelligent disassembly planning and decision-making

Disassembly process can be considered as an inverse process of assembly process, in which all fasteners and connectors among different subassemblies are eliminated. Most importantly, precedence constraints between disassembly operations must be satisfied to generate feasible disassembly sequences, which can be summarized as the disassembly sequence planning (DSP) problem. To explore the principles of DSP problems, disassembly modeling is an essential part for searching for the optimal or near optimal disassembly sequences regarding various products. The establishment of disassembly modeling is to express the relationship between disassembly process and disassembly components by using specific methods [58], which can be roughly divided into the following categories: graph-based methods, Petri nets as well as matrix-based methods. Graph-based methods, consisting of several typical methods like connection graph [59] and AND/OR graph (AOG) [60], give an intuitive way to describe precedence and associate information among the parts of an EOL product. Petri nets [61] give detailed geometrical and topological information on the components for system description and analysis, which are introduced to dynamically describe the disassembly process. Matrix-based methods [17,62] provide a more convenient means for computers to cope with disassembly constraints with the interference matrix method being the most common one. Based on the above disassembly modeling, a wide range of mathematical and heuristic strategies can be applied to find the optimal solutions from various alternatives. Mathematical methods [63–65] are initially utilized to search and identify the exact solution. Nonetheless, these mathematical approaches inevitably run into challenges with computational intractability as the problem size increases. Therefore in recent years, heuristics or metaheuristics [66–68] have been developed as efficient computation methods to generate near optimal solutions in order to deal with large scale or complex instances.

In addition to the traditional DSP, many scholars have conducted a lot of research on robot task planning in a dynamic disassembly environment. Robot task planning usually refers to selecting appropriate action and constraint sequences according to the initial state and target state of the manipulated object without interference, changing the state of the robot and the manipulated object, and realizing the transition from the initial state to the target state [69]. The research content of many scholars includes but is not limited to: motion planning and decision-making under uncertain conditions [70], real-time online task planning [71] and decision-making of intelligent robot manipulating objects, and separation planning and decision-making of target objects in complex environments [72], etc. Although such research results are very remarkable, they all have a common problem: the planner needs to input the symbolic state of the system, which usually needs to be pre-defined by symbolic experts, which limits the scope of application of task planning.

Many scholars have also conducted extensive research on human–robot collaboration (HRC), which includes collision avoidance and detection, motion planning under HRC environment, and the design of HRC systems [73]. These methods typically use multiple sensors to monitor the state changes between humans and robots, thereby achieving safe human–robot interaction in a dynamic environment [73,74]. More recently, some studies [75–78] suggest reinforcement learning (RL) methods to solve the stochastic and sequential disassembly decision-making problems owing to powerful interactive capabilities which have tangible benefits in terms of human–robot collaboration scenarios. In the above-mentioned HRC methods, the task and motion planning of robots generally lack interpretability.

To sum up, the above contents review the key technologies for building an intelligent disassembling system, these researches are gradually achieving the ultimate goal of building an intelligent disassembly system. Therefore, methods proposed in this paper aim to provide new ideas and a feasible solution for the intelligent disassembly of EVBs. The contributions of this paper are as follows:

1. Based on the manual disassembly process analysis of EVBs, an intelligent human–robot disassembling process and a framework of knowledge-driven flexible human–robot hybrid disassembly line are proposed for safe, efficient and flexible disassembly of EVBs.
2. Implementation of high-accuracy screw disassembly is proposed to replace the repetitive screw fastener operations, which is realized by designing a new disassembly actuator and adopting vision-based position and pose estimation and robust force-based screw insertion algorithms.
3. An autonomous disassembly planning system based on NeuroSymbolic is proposed to adapt to the highly dynamic and unstructured environment and correspondingly generate feasible disassembly primitive sequences.

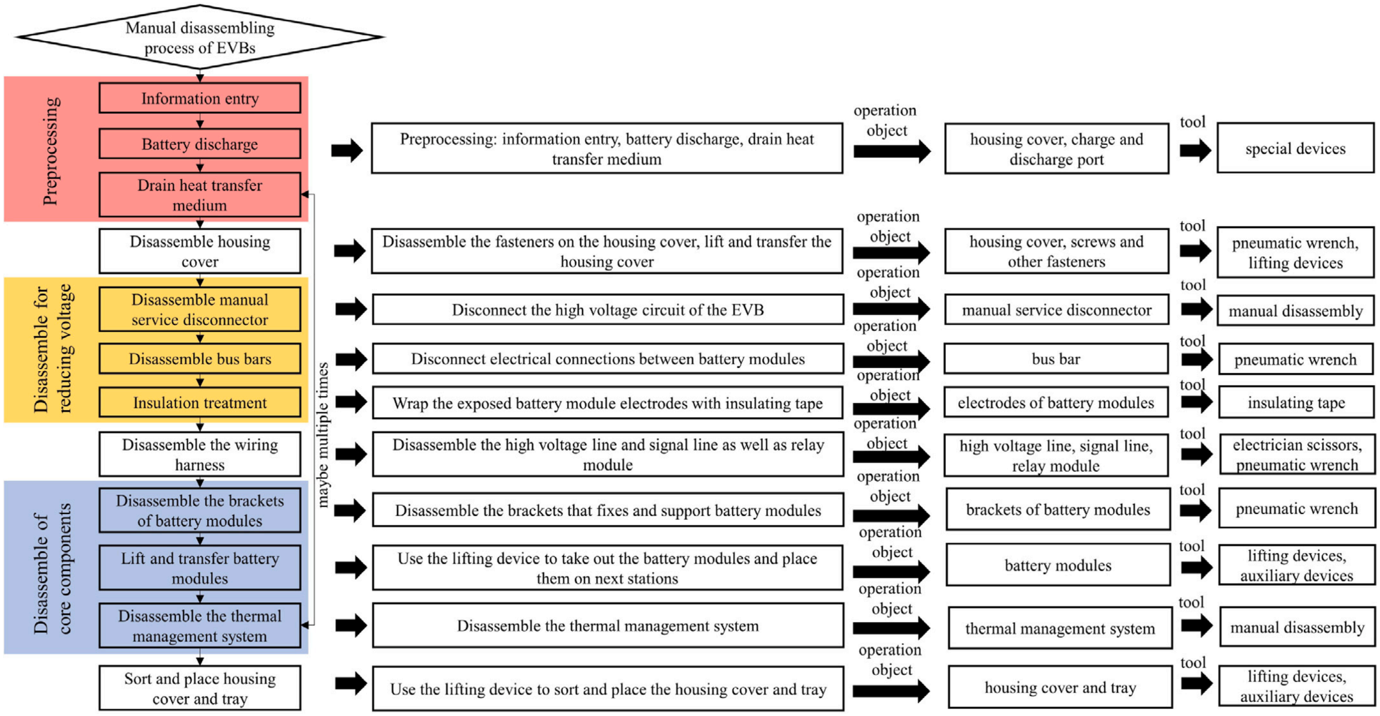

Fig. 1. Manual disassembly process of EVBs .

3. Research on intelligent disassembling process of EVBs

Most of previous researches [14,17] focused on the manual disassembling process of EVBs and investigations of automation potential. However, further researches are needed on human–robot disassembly and intelligent disassembly. Therefore, this section analyzes the disassembling process of EVBs and proposes a framework of knowledge-driven flexible human–robot hybrid disassembly line.

3.1. Manual disassembling process

By observing and summarizing the disassembling steps of the workers, we summarize the manual disassembly process of the EVBs (as shown in Fig. 1). We divided the disassembly of EVBs into the following main parts: preprocessing, disassembly for reducing voltage, disassembly of core components. Preprocessing is a necessary step that ensures the smooth operation of subsequent disassembling operations and protects the life safety of the workers on the disassembly line. The disassembly for reducing voltage can not only disassemble the components, but also further reduce the voltage on the basis of the preprocessing operations. Disassembly of core components is mainly for the battery modules and related components, which have the greatest recycling value. After the disassembly of core components, the disassembly of the entire EVB is basically completed.

3.2. Intelligent disassembling process analysis

3.2.1. Links that are easy to intelligently disassemble

Disassembly of screws and other fasteners. According to the manual disassembly process of EVBs, the disassembly of screws and other fasteners is generally performed by workers using a pneumatic wrench. On the one hand, compared with operations performed manually, it is much easier to realize intelligent disassembly. On the other hand, it is unscrewing operations that take up approximately 40% of all disassembly activity [16,17]. Replacing the manual disassembly of screws and other fasteners with robotic intelligent disassembly will greatly reduce the workload of workers and improve the efficiency of the disassembly process of EVBs.

In order to realize the intelligent disassembly of screws and other fasteners, high-accuracy position and pose estimation of screws and other fasteners, robust insertion based on force perception and relevant robotic disassembly actuators need to be further investigated.

Lifting and transferring battery modules. During the process of lifting and transferring the battery module, workers mainly perform two operations: securing lifting straps (other similar devices) and the battery module to each other, guiding the lifted battery modules and stacking them on a suitable station. As for the first operation, it is human adaptability that is fully utilized to secure various battery modules with lifting devices. And the reason why the second operation is performed manually is that the traditional lifting equipment cannot automatically control the trajectory and landing point of the lifted heavy objects.

The above operations are easy to realize by robotic intelligent disassembly. More specifically, automatic lifting and fixing devices, which are suitable for various battery modules, need to be designed. And industrial robotic arms are utilized to move and stack battery modules with machine vision algorithms.

3.2.2. Links that are difficult to intelligently disassemble

Disassembling operations performed manually. These operations include the insulation of battery module electrodes, the release operation of the snap fit of the plastic shell at both ends of the bus bar and the disassembling operations of the wiring harness, especially the harness plug, etc. These operations, even when performed manually, require many attempts and great dexterity to succeed. Furthermore, most of these operations will be carried out in a destructive way for the efficiency of disassembly. It is unrealistic for these operations to be implemented intelligently in the short term.

There are two main reasons that hinder the realization of intelligent disassembly of these operations:

1. The operation objects involve deformable objects, such as the insulating tape and the wiring harness.

2. The connections and assembly in these operations are difficult to release, which requires great dexterity, such as disassembling the harness plug and the snap fit of bus bars.

3.2.3. Intelligent human–robot disassembling process

Based on the above analysis, we propose an intelligent human–robot disassembling process (As shown in Fig. 2). Parallel mode ensures high efficiency of disassembly. It is worth noting that the human–robot disassembling process puts forward higher requirements for the autonomous control of disassembling system, so that it has sufficient autonomy to adapt to highly dynamic scenarios and has sufficient reliability to ensure the personal safety of workers.

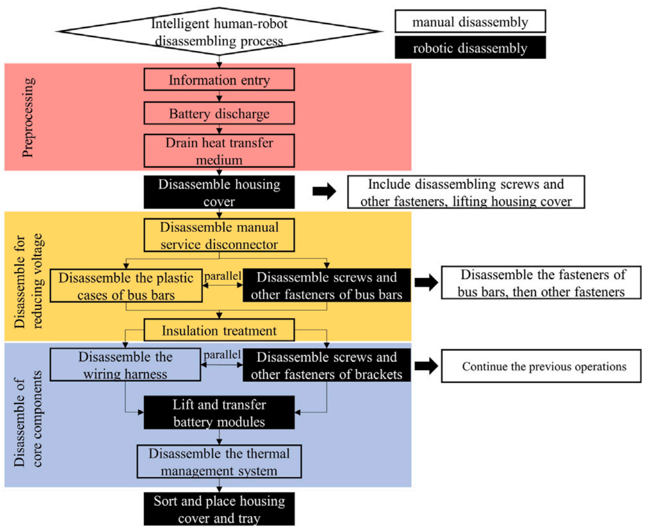

Fig. 2. Human–robot disassembly process of EVBs .

3.3. Framework of knowledge-driven flexible human–robot hybrid disassembly line

Based on the problems and challenges unique to the disassembling scenarios of EVBs, we propose a framework for a novel knowledge-driven flexible human–robot hybrid disassembly line (as shown in Fig. 3).

3.3.1. Introduction of disassembly line

Different from the traditional fixed assembly line, the flexible disassembly line is composed of a series of independent workstations, each of which has different operating capabilities and is represented by various capability attributes. For example, there is a workstation with a lifting load of 20 kg, which can complete the operation of disassembling screws additionally, while another workstation can complete the operation of removing adhesive, and the lifting load is 300 kg at the same time. Furthermore, the flexible disassembly line can split the overall EVB disassembling task layer by layer to the primitive-level subtasks based on knowledge, and intelligently generate the optimal disassembling primitive sequence that can complete the corresponding subtask. Combined with the different operating capabilities of each workstation on the disassembly line, the batteries to be disassembled are allocated and scheduled to flow among the workstations. In this way, the disassembling tasks of EVBs with different conditions are completed independently and efficiently, aiming at the maximum benefit for subsequent recycling and remanufacturing. In view of the fact that dexterous operation cannot be achieved autonomously by robots in the short term, some tasks requiring high flexibility are still completed manually. Moreover, considering that mechanical disassembly of EVBs depends on a large amount of established knowledge and will accumulate a wealth of new knowledge in practice, we adopt a knowledge-driven approach and end up with a flexible human–robot hybrid disassembly line.

3.3.2. Operation flow of disassembly line

As shown in the disassembly planning module in Fig. 3, When a EVB enters the disassembly line, the system can identify the type, model and other information of the battery by scanning the QR code or other special signs. The information acquired can help the system to subdivide disassembling tasks into subtasks at the knowledge level. According to the type of battery, such as a ternary lithium battery or a lithium phosphate battery, a blade battery or a magazine battery, the system can generate a preliminary task sequence. We call the preliminary task sequence as first-level subtask, which is not a specific executable action sequence. In particular, the first-level subtasks are generated based on the ontology of the disassembling process in the knowledge base, and clearly define the disassembling goal and the procedure of disassembling process. In addition, some inspection methods (increase the reusability rate of EVBs) and safety measures (reduce the rate of disassembling accidents) are also introduced and executed. The above operations provide direction for the refinement of the disassembling tasks in the next step.

Next, the system refines first-level subtasks into second-level subtasks according to the technical specifications of EVBs. Since different types of batteries will vary in many aspects such as specific size, battery capacity and the number of battery cells, the required parameters in the second-level subtasks are further generated and refined based on corresponding information in the knowledge base. For example, when disassembling a certain specific battery, the system needs to disassemble 16 screws before the lifting operation of the upper cover can be completed. After the removal of the bus bars and auxiliary brackets of a certain number, the residual energy of the battery module can be detected, which is utilized to determine whether to continue disassembling for cascade utilization or material recovery. Combined with the generated second-level subtasks and the capabilities of each workstation pre-stored in the knowledge base, we can assign the corresponding disassembling tasks to the corresponding workstations, thus determining the flow direction of the battery in the disassembly line.

The above two steps are task decomposition completed in the knowledge domain, but the end-of-life EVBs will not be totally consistent with the knowledge in the knowledge base due to the fact that certain parts may be corroded, damaged or even missing. In this case, we specially designed an autonomous system within the framework of NeuroSymbolic. With the help of the knowledge base storing disassembling primitives and predicates and the task planning system based on NeuroSymbolic, the system autonomously generates sequences of disassembling primitives related to the disassembling subtask. In particular, neural predicates are creatively introduced into the task planning system, which can automatically map sensor inputs into knowledge symbols and then determine the optimal execution sequence based on the logical search method. Such a design ensures that the task planning system can smoothly convert the secondary subtasks into an execution sequence, adjusting in real time according to the actual situation perceived by the sensor.

Our proposed disassembly line has the following advantages.

1. The flexible human–robot disassembly line based on knowledge-driven hierarchical planning can obtain near-optimal results of disassembly planning with a relatively small computational load.
2. The knowledge-driven disassembly planning enables the system to continuously expand the original knowledge base and perform reasoning based on the knowledge base, which enhances the ability of the system to flexibly handle uncertain events in the disassembly process.

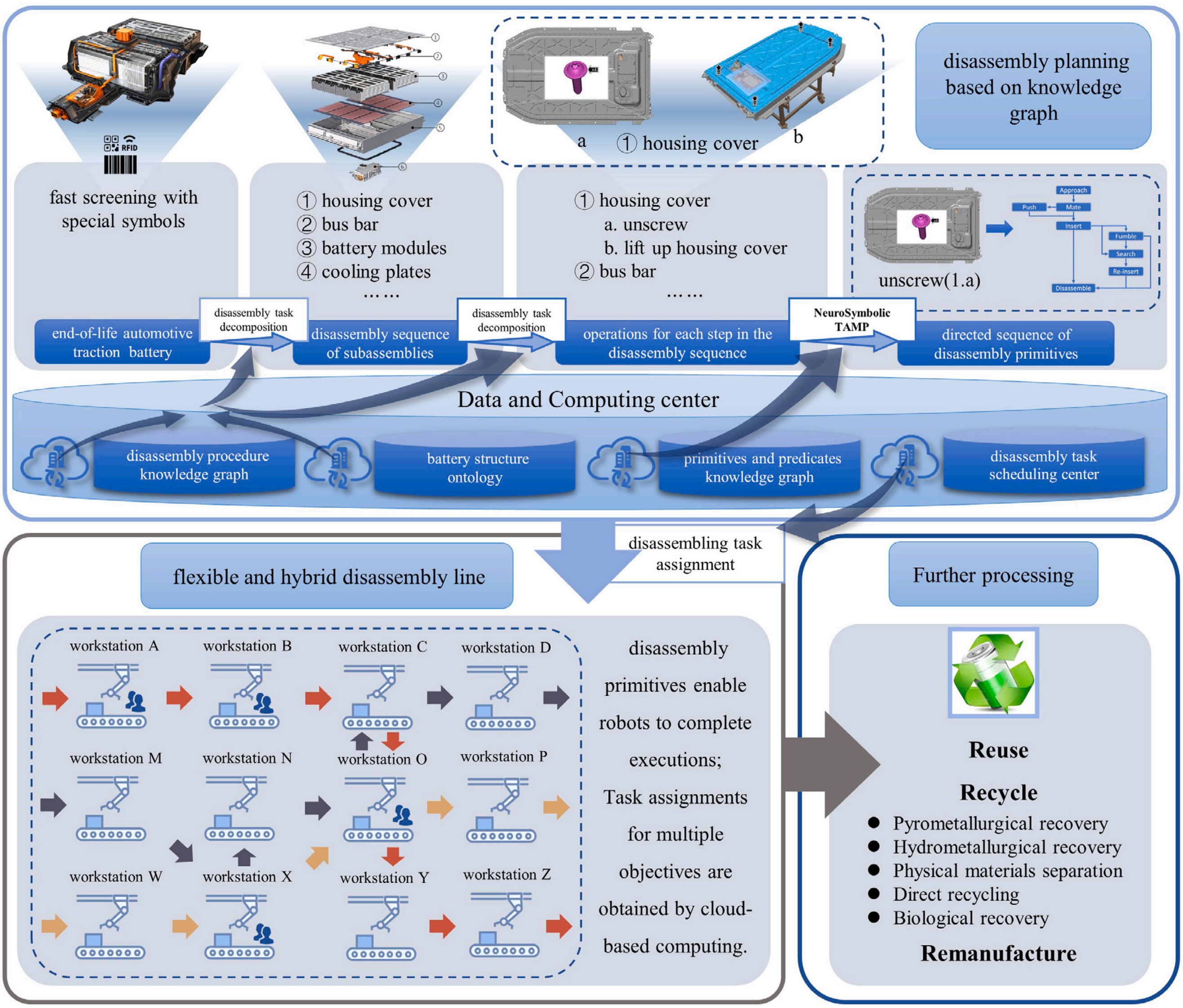

Fig. 3. Overall framework of the knowledge-driven flexible human–robot hybrid disassembly line.

3. The task planning system based on NeuroSymbolic has the advantages of autonomy, interpretability, learnability, and scalability. It should be noted that interpretability not only ensures the safety of human–robot disassembly, but also allows workers to intervene technically in unexpected situations.

Our implementation in this paper focuses on key technologies involved in the above-mentioned disassembly line, including implementation of high-accuracy screw disassembly and task and motion planning based on NeuroSymbolic. Experiments show that our system can effectively complete the screw disassembling tasks in complex disassembling scenarios.

4. Key technologies

4.1. Implementation of high-accuracy screw disassembly

In order to achieve real-time and high-accuracy operations, we apply the YOLOv5 for object detection, plane fitting based on Random Sample Consensus (RANSAC) [79] for pose estimation, and introduce Kalman filtering (KF) [80] to correct the results. Furthermore, when performing insert operation, the most critical step in the process of disassembling fasteners, we adopt a combination of active and passive compliance to compensate for the errors caused by visual position and pose estimation. For one thing, we developed a new disassembly actuator for passive compliance at hardware level (as mentioned above). For another, a Markov decision process (MDP) model based on force perception is built for implementation of robust insertion. The above-mentioned algorithms and disassembly actuator are essential for high-accuracy screw disassembly operations.

4.1.1. Disassembly actuator

In order to make the robot efficiently complete the screw disassembly task, we designed a passive and compliant pneumatic torque actuator with vision sensing (as shown in Fig. 4), including a pneumatic power subject, a square shaft transmission limit module, a rod limit module and a vision module.

The structure and function of the pneumatic power subject is similar to that of the pneumatic impact wrench, including cylinders, blades, rotors, impact hammers and other parts. The transmission square shaft limit module includes a shaft limit shell, a transmission square shaft, a sliding bearing, a spring and a square shaft bushing, which is responsible for transmitting the torque output by the pneumatic power subject to the long rod. At the same time, with the help of the spring, the transmission square shaft and the long rod can move freely within

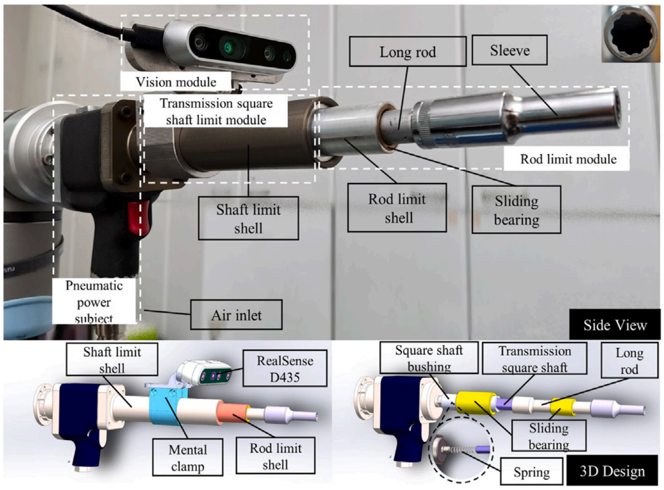

Fig. 4. (Top) The structure of real machine. (Bottom) Internal structure of the disassembly actuator.

a certain range in the axial direction, which meets the passive compliance requirements of the disassembly process. More specifically, this structure can not only compensate for the error caused by depth measurement during positioning, but also prevent the rigid collision of the loosening action of the screw and the disassembly actuator when the screw is being disassembled.

The rod limit module includes a rod limit shell, a sliding bearing, a long rod and a sleeve. The long rod ensures that the actuator can disassemble screws that are difficult to reach, and enhances the adaptability to complex disassembly environments. The assembly of the sleeve and the long rod is detachable, which is convenient for replacing other types and sizes of sleeves. The vision module includes a metal clamp and an Intel RealSense D435 camera. With the D435 rigidly assembled on the shaft limit shell through the metal clamp, the relative position of the target screw and the front end of the disassembly actuator can be clearly observed in the field of view of the D435 camera, which is conducive to the accurate judgment and perception of neural predicates. Mental clamp design ensures firm assembly and compact structure.

Under complex disassembling conditions, the mechanism related to the spring can play a passive and compliant role in improving the success rate of sleeve clamping. For the hexagonal screws studied in this paper, we also specially used a 12-point sleeve, which is convenient to make the sleeve successfully clamped under a small rotation angle. For this reason, in the experiment of the real scene, our disassembly actuator will keep rotating around its own axis at a slight angular velocity while descending to perform the **Insert** primitive.

4.1.2. High-accuracy screw position and pose estimation based on visual perception

Due to the complex and various end-of-life status of waste EVBs, the screws and fasteners to be disassembled will be in random and uncertain states. Therefore, the premise for the successful disassembly of screws and fasteners is the fast and high-accuracy detection of the position and pose of targets. In the subsection of small fastener object detection discussed in the previous literature review, relevant researches not only failed to detect screws and other fasteners in real time, but also failed to perform pose estimation on these fasteners. Furthermore, in the real physical world, there are many deviations due to unfavorable factors such as invalid depth information, surface wear, tilt, occlusion and so on. Hence, we redesign the procedure for position and pose estimation to solve these challenges.

The overall framework of high-accuracy position and pose estimation based on vision perception is shown in Fig. 5. Firstly, the robot adjusts the disassembly actuator to the coarse position and pose near the target fastener before performing object detection and obtaining the bounding box and target center based on the RGB information of the visual sensor. Next, 3D point cloud of the adjacent plane is generated based on the bounding box and depth information, and the normal vector of the adjacent plane, which can be transformed to the pose of the target fastener, is obtained by the plane fitting algorithm based on RANSAC. Finally, the Kalman filter is utilized to improve the accuracy of 6D position and pose estimation. When the covariance of process noise δ is less than the preset threshold ϵ, the latest result is output as the final result. Otherwise, the robot adjusts the disassembly actuator to the latest position and pose and repeats the above process. The key implementation details are elaborated as follows.

YOLOv5 for object detection. The latest one-stage detection network YOLOv5 is selected to perform the screw detection tasks owing to the high accuracy and fast reasoning speed of the YOLO network. The detection model is trained in a supervised process on a set of 2500 labeled images, which are sourced from eight typical types of EVBs, simultaneously using data augmentation to extend the amount of different images. The dataset contains two categories of fasteners: hex screw and hex nut, wherein the hex screws account for the 80% of the total number of fasteners. As for the three models of YOLOv5 network, YOLOv5-S stands out after our investigations with mAP_{75}87.56%. YOLOv5 detector training hyperparameters are as follows: learning rate 1×10^{-4}, batch size 10, input image size $416\times416\times3$, train for 100

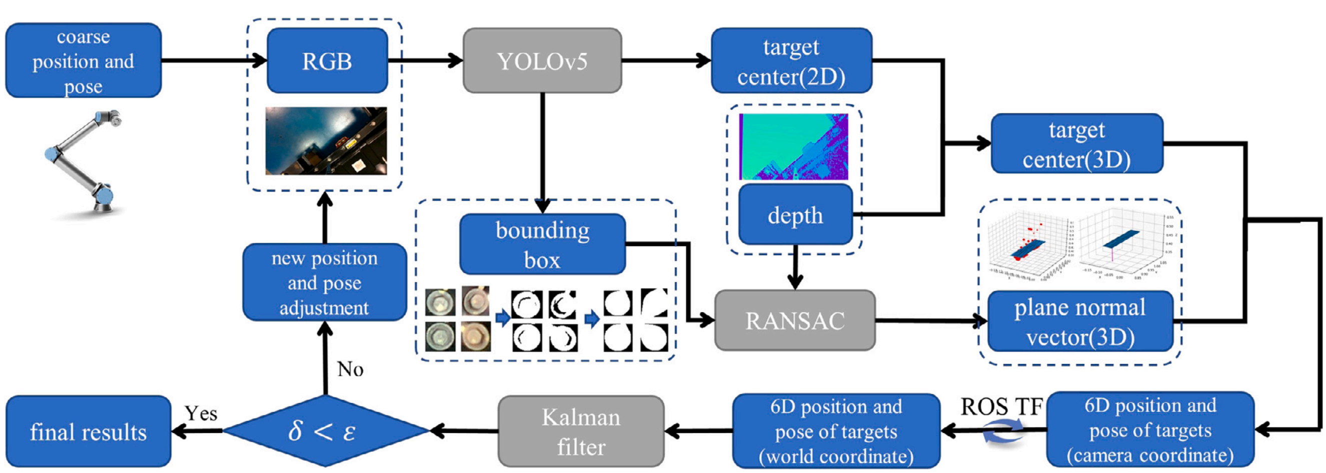

Fig. 5. Overall framework of high-accuracy position and pose estimation based on vision perception.

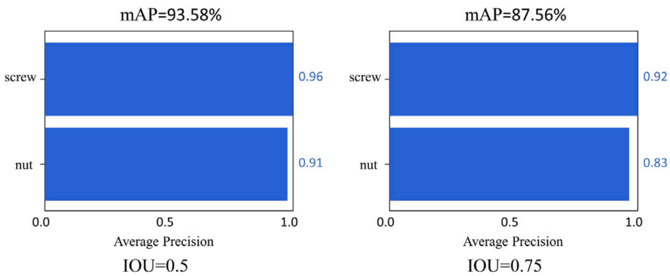

Fig. 6. AP and mAP of different IOU by YOLOv5 object detection algorithm.

epochs. Average Precision (AP) and mean Average Precision (mAP) of different IOU by YOLOv5 is shown in Fig. 6. Despite the sharp decrease of mAP when IOU is 0.9, the performance of the YOLOv5 detector is still good enough.

We set the center (x_c, y_c) of the detected fastener's bounding box as the center of fasteners due to the fact that bounding boxes are generally tightly circumscribed with fasteners. Furthermore, the viewing angle of the camera is often perpendicular to the fastener (results of pose estimation), which can further reduce the error caused by the above design.

Plane fitting based on RANSAC for pose estimation. Particularly, some problems will be encountered in the pose estimation process of screws and other fasteners. On the one hand, the small size of fasteners increases the difficulty of sensor observation. On the other hand, fasteners are generally assembled into parts and only expose partial entity of the head. These factors make traditional pose estimation algorithms [81] unsuitable for pose estimation of fasteners. Therefore, we apply the plane fitting based on RANSAC and morphological operations for fastener pose estimation.

Firstly, how to select the appropriate adjacent planes for plane fitting becomes the key factor to ensure the accurate pose estimation of fasteners. For one thing, the selected adjacent plane is supposed to be as close as possible to the target fastener. For another, the area of the selected adjacent plane should be reasonable enough not to be too big to achieve accurate plane fitting results or be too small to obtain enough point cloud data causing the failure of plane fitting.

Based on the above selection principle of the adjacent plane, we use the bounding boxes generated by the YOLOv5 object detection algorithms as the boundary of the adjacent planes. Moreover, in order to distinguish the adjacent areas in bounding boxes, extra operations of Otsu thresholding [82] and morphological convex hull [83] are implemented to segment the target fasteners completely(as shown in Fig. 7). The selected adjacent planes are the regions of black background.

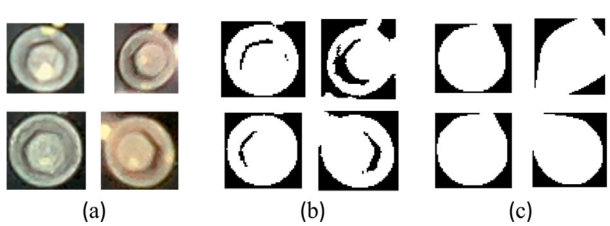

Fig. 7. Image processing operations. (a) original images from bounding boxes. (b) binary images after Otsu thresholding. (c) binary images after convex hull operations.

Then the point cloud (3D points), which is generated based on the selected adjacent planes, is utilized for plane fitting algorithm based on RANSAC to robustly obtain the normal vectors of fitting planes. The normal vectors can be converted into poses of fasteners by the following method(the method is not unique).

The normal vector of fitting plane is known as $\vec{n}$(in camera coordinate). And the direction vector of the z-axis positive direction of the camera coordinate is $[0 \quad 0 \quad 1]^T$, the unit cross product of both vectors is as follows.

$$\vec{n}_1 = \frac{\vec{n} \times [0 \quad 0 \quad 1]^T}{|\vec{n} \times [0 \quad 0 \quad 1]^T|} \tag{1}$$

Then we can obtain the unit cross product of normal vector $\vec{n}$ and vector $\vec{n}_1$.

$$\vec{n}_2 = \frac{\vec{n} \times \vec{n}_1}{|\vec{n} \times \vec{n}_1|} \tag{2}$$

According to the nature of the cross product, the vectors of $\vec{n}$, $\vec{n}_1$, $\vec{n}_2$ are the pairwise orthogonal unit vectors. Therefore, by combining these vectors, orthogonal matrix R is obtained.

$$R = [\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}]^T \tag{3}$$

Through mathematical calculation, it is obvious that the following equation can be obtained.

$$R \times \vec{n} = [0 \quad 0 \quad 1]^T \tag{4}$$

Therefore, matrix R is the rotation transformation from the normal vector of fitting plane to the direction vector of the z-axis positive direction of the camera coordinate. That is to say, matrix R is the pose of target fastener.

In sum, the plane fitting algorithm based on RANSAC uses the point cloud based on the selected adjacent planes as input and calculates the normal vectors of fitting planes. Subsequently, the normal vectors are converted into poses of fasteners. In this way, we can obtain the pose of fasteners efficiently and robustly.

Kalman filtering to reduce deviation. Since there are various factors in the real environment that affect the quality of the RGB image and depth information collected by the camera, the reliability of the estimated screw position and pose is often very limited. Considering that the target screw is stationary during the positioning phase, the estimated value of its position and pose can be regarded as a constant true value mixed with Gaussian noise. By introducing Kalman filtering [80], which is widely used in object tracking, an iterative positioning method with repeatedly sampling is established to correct the estimated screw position and pose. In the filter, the estimated screw position and pose $\hat{x}$ is expressed by the center coordinates $\hat{x}^p$ and normal vector $\hat{x}^o$ of the upper surface:

$$\hat{x} = (x^p, y^p, z^p, x^o, y^o, z^o) \tag{5}$$

For the kth iteration,

$$\hat{x_k}' = \hat{x_k} + K'(z_k - H_k \hat{x_k}) \tag{6}$$

$$P_k' = P_k + K' H_k P_k \tag{7}$$

$$K = H_k P_k H_k^T (H_k P_k H_k^T + R_k)^{-1} \tag{8}$$

where P_k is the covariance matrix, H_k is the identity matrix, R is the covariance matrix (representing the sensor noise), and P_0 is the initial covariance estimation matrix.

This iterative method involves an adaptive adjustment strategy, that is, the disassembly actuator will adjust its posture according to the results of each iteration so that the sleeve on the disassembly actuator is aligned with the expected fastener. The aim is to perform object detection with the frontal image of the screw to reduce errors caused by screw tilt.

4.1.3. Robust insertion based on force perception

Since the "Insert" operation relies on extraordinarily high positioning and control accuracy, in addition to minimizing the visual positioning error, a strategy combining both active and passive compliance is adopted. A customized disassembly actuator (as shown in Fig. 4) and an insertion decision model based on force perception are developed to make up for the residual error.

During the insertion process, when the disassembly actuator descends close enough to the screw, the deviation caused by the positioning results and the movement of the robotic arm cannot be corrected by visual means. To handle such situations with restricted visual perception and control accuracy, a Markov decision process (MDP) model driven by active exploration of contact state based on force perception is established as a scheme to achieve robust insertion of screws and other fasteners (as shown in Fig. 8).

We define the contact state criterion, and divide the contact state when the sleeve and the screw are close to each other into three types:

1. Successfully clamped ($s_t = success$): the inner wall of the sleeve completely surrounds the side of the screw, which is regarded as a complete clamping.
2. Partially clamped ($s_t = part$): the lower edge of the sleeve is in contact with the upper surface of the screw and is thus under relatively larger pressure, which is regarded as a partial clamping.
3. Not clamped ($s_t = lost$): the sleeve is in direct contact with the surface of the battery pack, which is regarded as totally not clamped.

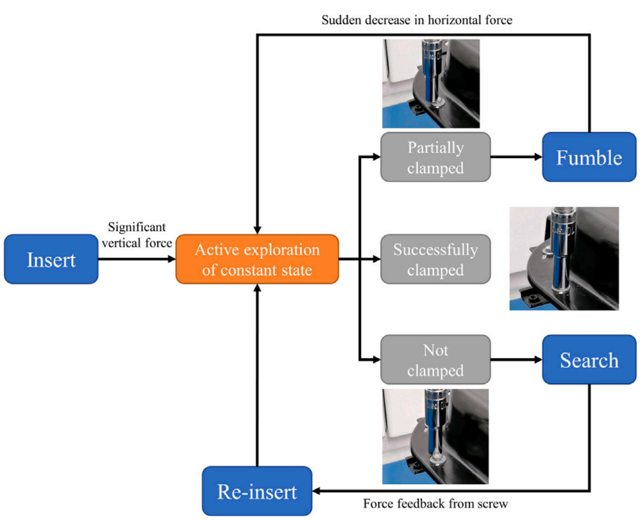

Fig. 8. MDP model based on force perception and contact state illustration.

Hence, the continuous insertion process can be represented by a discrete state sequence $s(n)$:

$$s(t) = (s_0, s_1, \dots, s_t, \dots, s_G) \tag{9}$$

where $s_0 = init$ is the initial state. In this model, each action decision a_t is generated based on the current state s_t to approach the goal state $s_G = success$.

$F(n)$ composed of the force feedback for the last n moments is used to describe the current contact state. Once a sudden change occurs and thus $F(n)$ satisfies certain features, it indicates that the contact state is likely to have changed, which is also in line with the expectation of the primitive function.

As shown in Fig. 8, active exploration will be performed in this case to determine the latest contact state, which involves a series of slight translations in its horizontal directions for more detailed force feedback. When performing exploration, if $s_t = success$, there will be obvious force feedback in opposite horizontal directions. Otherwise, the vertical force of $s_t = lost$ will be significantly reduced compared with $s_t = part$. Our contact state criterion is modeled based on these force feedback features combined with experimental data.

As for "Re-insert", an obvious lateral force is sensed during the execution of the "Search", and the repositioning of the screw based on force feedback is realized according to the direction of the force and the identified screw radius. It can be found that under the guidance of this decision model, the robot will make constant insertion and adjustment attempts until the screw is successfully clamped.

4.2. PDDL-based NeuroSymbolic TAMP

In order to ensure high-accuracy disassembly, the robot needs to perform many actions, including positioning and alignment, etc. Besides, changes in the environment call for some different operations, for example, when there are obstacles, they need to be removed in advance. The application of these actions needs to be scheduled according to the actual situation. Therefore, we proposed an architecture of NeuroSymbolic task and motion planning (NeuroSymbolic TAMP) [9] to adapt to highly dynamic disassembling scenarios. This section provides a detailed framework of our intelligent disassembly system, including the introduction of NeuroSymbolic, the action primitives, the neural predicates, the TAMP based on NeuroSymbolic and the future vision.

4.2.1. Introducing the NeuroSymbolic

On the premise of obtaining the basic action primitives, PDDL is introduced to model the disassembly task for logical reasoning. However, it is obviously infeasible to symbolize all detailed disassembly states due to the complexity of disassembling scenarios, so we introduce neural predicates that extract discrete symbols from continuous visual inputs using probabilistic learning methods for describing the task-relevant disassembly states. In this way, high-level reasoning and low-level perception are organically combined for autonomous and explainable disassembly of EVBs.

During the experiments, we found that two states great-ly affect the success rate of the disassembling, involving the final results of position and pose estimation, obstacles around the target respectively. The former state occurs when the calculated results have great errors with true values, the latter state occurs when the target is covered by obstacles (mostly disassembled fasteners thrown off by the previous disassembling operation), both resulting in the failure of the disassembly. Therefore, we introduce two neural predicates ***target_aim()*** and ***target_clear()*** to indicate whether the sleeve is aligned with the screw and whether there are obstacles hindering disassembly above and near the screws, which enhance the reliability of disassembling and the ability to deal with complex working conditions. Moreover, a force-based decision model is introduced for the robust insertion operation to improve the success rate of the screw disassembling process.

4.2.2. Action primitives

With further observations on manual disassembly of industrial workers, we may consolidate the operational experience into several core action primitives, which enable the workers to complete the disassembling work efficiently.

Approach: This primitive manages the robot's disassembly actuator to move close to the target for subsequent actions.

Mate: This primitive adjusts the robot arm's disassembly actuator to get its central axis aligned with the target(screw)'s central axis.

Push: If perception indicates that there is not enough space for the disassembly actuator to conduct disassembly, this primitive will be invoked to push away obstacles around the target.

Insert: This primitive directs the disassembly actuator to descend along the screw axis with a small angular velocity so that the sleeve and the screw have valid contact.

Fumble: This primitive is executed when the current state is ***part_clamped*** after **Insert** operation. *i.e.*, the disassembly actuator is not perfectly matched the target(screw) and more adjustments of disassembly actuator are required.

Search: If the location of **Insert** operation has large deviation with the location of the target(screw), the current state of agent is neither ***clamped*** nor ***part_clamped***. The disassembly actuator is directed to

Primitive	Pre-condition	Result	Input Sensor	PDDL Definition	Function Description
Approach	1. there is a coarse pose of the screw to be disassembled acquired via visual perception 2. the sleeve is not near above the screw 3. the sleeve is not close to the screw	1. the sleeve is near above the screw 2. the sleeve is aligned with the screw 3. there are no obstacles near the screw	RGB-D camera	(:action Approach :param (coarse_pose camera) :pre (and(have(coarse_pose)) (not(above_screw)) (not(near_screw))) :eff (and(above_screw) (target_aim(camera)) (target_clear(camera)))	This primitive manages the robot's disassembly actuator to move close to the target for the subsequent actions.
Mate	1. the sleeve is near above the screw 2. the sleeve is not aligned with the screw	1. the sleeve is aligned with the screw	RGB-D camera	(:action Mate :param (camera) :pre (and(above_screw) (not(target_aim(camera)))) :eff target_aim(camera))	This primitive adjusts the robot arm's disassembly actuator to get its central axis aligned with the target's central axis.
Push	1. the sleeve is near above the screw 2. there are obstacles near the screw.	1. there are no obstacles near the screw 2. the sleeve is aligned with the screw	RGB-D camera	(:action Push :param (camera) :pre (and(above_screw(camera)) (not(target_clear(camera)))) :eff (and(target_clear(camera)) (target_aim(camera)))	If perception indicates that there is not enough space for the disassembly actuator to conduct disassembly, this primitive will be invoked to push away obstacles around the target.
Insert	1. the sleeve is near above the screw 2. the sleeve is aligned with the screw 3. there are no obstacles near the screw.	1. the screw is clamped to the sleeve 2. the sleeve is close to the screw 3. the sleeve is not near above the screw.	RGB-D camera force sensor	(:action Insert :param (camera force_sensor) :pre (and(above_screw) (target_aim(camera)) (target_clear(camera)) :eff (and(clamped(force_sensor)) (near_screw) (not(above_screw)))	This primitive directs the disassembly actuator to descend along the screw axis with a small angular velocity so that the sleeve and the screw have valid contact. .
Fumble	1. the sleeve is close to the screw 2. the screw is partially clamped to the sleeve	1. the screw is clamped to the sleeve.	force sensor	(:action Fumble :param (force_sensor) :pre (and(near_screw) (part_clamped(force_sensor)) :eff (clamped(sensor))	This primitive commands the disassembly actuator to fumble around with a spiral trajectory in order to complete clamping if the screw is partially clamped.
Search	1. the sleeve is close to the screw 2. the screw is totally not clamped to the sleeve 3. there is not an estimated pose of the screw acquired via force perception.	1. there is an estimated pose of the screw acquired via force perception.	force sensor	(:action Search :param (tactile_pose force_sensor) :pre (and(near_screw) (not(clamped(force_sensor))) (not(part_clamped(force_sensor))) (not(have(tactile_pose)))) :eff (have(tactile_pose))	This primitive commands the disassembly actuator to search around for the screw with a spiral trajectory if the screw is totally not clamped.
Re-insert	1. there is an estimated pose of the screw acquired via force perception.	1. there is not an estimated pose of the screw acquired via force perception 2. the screw is clamped to the sleeve.	force sensor	(:action Re-insert :param (tactile_pose force_sensor) :pre (have(tactile_pose)) :eff (and(not(have(tactile_pose))) (clamped(force_sensor)))	The disassembly actuator executes the ***Insert*** operation again based on the new tactile pose of the target.
Disassemble	1. the screw is clamped to the sleeve 2. the screw is not disassembled.	1. the screw is disassembled	RGB-D camera force sensor	(:action Disassemble :param (camera force_sensor) :pre (and(clamped(force_sensor)) (not(disassembled(camera)))) :eff (disassembled(camera))	This primitive directs the disassembly actuator to rotate counter-clockwise to loosen the target (screw).

Fig. 9. Description of disassembly primitives.

search around the adjacent areas to get the new tactile pose of the target.

Re-insert: The disassembly actuator executes the **Insert** operation again based on the new tactile pose of the target.

Disassemble: This primitive directs the disassembly actuator to rotate counter-clockwise to loosen the target (screw).

The final description of disassembly primitives is represented in Fig. 9. Based on the above description, the disassembly planning process based on logic reasoning is shown in Fig. 10. The disassembling planning system can generate the proper sequence of disassembling primitives according to the current state, which is symbolized by neural predicates. Benefiting from the neural predicates, the pre-condition can be checked before the execution of each primitive. If the current state is inconsistent with the predicted state, the disassembling planning system will be recalled to generate the new primitive sequence. In the corner case, the primitives might be executed multiple times due to limited execution effects or inaccurate sensing. This method ensures the disassembly planning system can support the dynamic and unstructured environment efficiently.

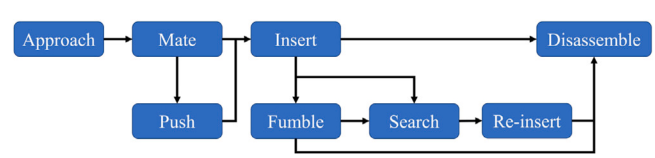

Fig. 10. Disassembly planning process based on logic reasoning.

4.2.3. Neural predicates

Compared with our previous paper [9] which focuses on the implementation of neural predicates in the simulation environment, we continue to use pre-trained VGG-16 [84] for the judgment of disassembling status in real scenes. 1200 images equally divided into four categories are utilized to train the VGG-16 model (as shown in Fig. 11). The four-category problem achieves an accuracy of 97.89% after 40 epochs of training. Such accuracy is sufficient for subsequent reasoning tasks. Besides the ***target_aim()*** and ***target_clear()***, we can regard the proposed force-based decision model as neural predicates to

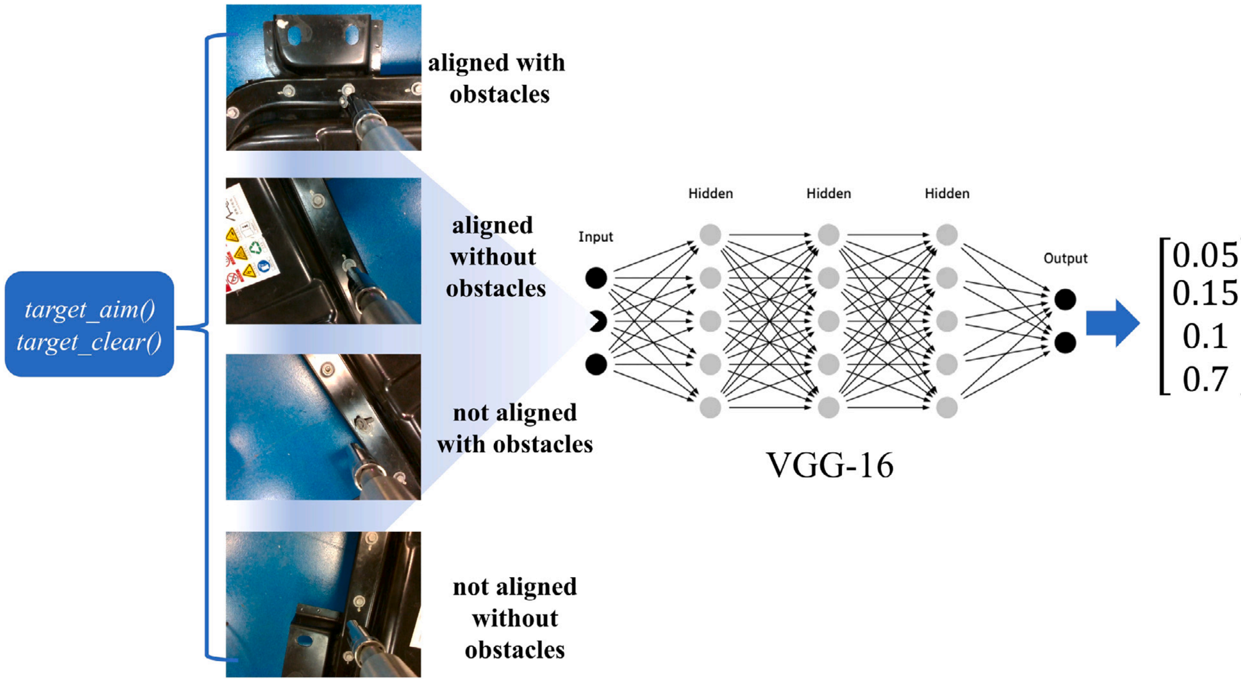

Fig. 11. Neural predicates for screw disassembly.

some extent, which both convert high-dimensional sensor information into logical symbols.

It is worth noting that we provide context for the perception module of the neural predicates through high-level symbolic reasoning. The perception module only needs to recognize and abstract within a limited range according to the current context, which greatly reduces the training difficulty of the neural network. The introduction of neural predicates in NeuroSymbolic TAMP enables the system to update the state according to dynamic disassembling scenarios and perform task planning at any time instead of relying on artificially abstracted and unchangeable states in advance.

4.2.4. TAMP based on the NeuroSymbolic

Benefiting from the disassembly primitives and neural predicates defined above, we use PDDL to abstract the complex planning problem in continuous space, and convert task planning to a process of finding optimal action plans in symbolic space. Considering the low complexity of the primitive space and low computational cost, the First Input First Output (FIFO) algorithm based on breadth-first traversal search [9] is used to select the one with the highest probability of success among the shortest steps from all the obtained action sequences, given the original state S_0 and the goal state S_G.

4.2.5. Future vision

In the future, in order to deal with more complex disassembly tasks of EVBs in dynamic environments, on the one hand, we consider constantly introducing new predicates to describe the state in as much detail as possible, and designing more primitives to provide more diverse means of disassembly (such as destructive disassembly) for deeper disassembly of EVBs. On the other hand, the continual learning of neural predicates is another focus of the following research, which can constantly improve the accuracy of neural predicates and adapt to the new changes in the environment.

5. Experimental configurations

Based on the actual disassembling scenes of EVBs, an intelligent disassembling workstation (as shown in Fig. 12) was built for experiments and tests, which is mainly divided into four parts: a disassembling workstation, a waste EVB, a 6-DOF UR10e robot and a disassembly actuator. The waste EVB is placed stably on the disassembling workstation and is the disassembling object for verifying our theory and algorithms. Moreover, we specially choose the UR10e robot with high payload (12.5 kg) and long working radius (1300 mm). A built-in torque sensor in the last joint of UR10e robot provides force input during the disassembling process. Most importantly, a specially designed disassembly actuator is mounted on the end joint of UR10e robot. In addition to the disassembly actuator for screw disassembly tasks, more disassembly actuators suitable for different disassembling tasks will be designed with further research.

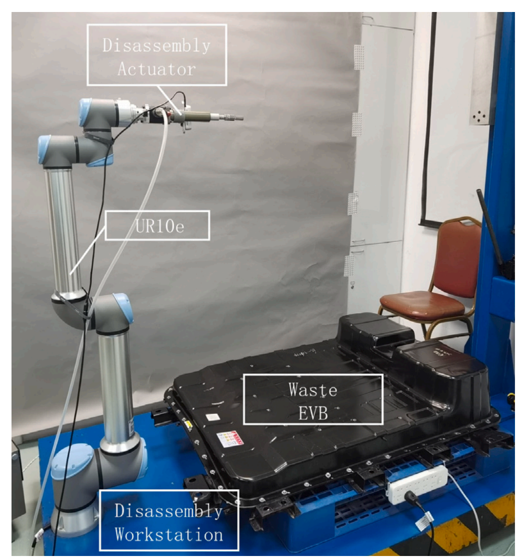

Fig. 12. Experimental disassembly platform.

Precision 5820 Tower X-Series from Dell is chosen as the computing platform for running the algorithm. The specific parameters for computing hardware and software are shown in the Table 1.

Table 1
Parameters for computing hardware and software.

Software & hardware	Parameters
CPU	Intel i9-10900X 3.70GHz
GPU	GeForce RTX 3080
RAM	128GB
Operating System	Ubuntu 18.04.6 LTS
Robot Operating System	ROS Melodic
Programming Language	Python
AI Framework	Pytorch

5.1. Vision-based position and pose estimation and neural predicates

To verify the accuracy of the position and pose estimation based on visual perception, an Ablation experiment was conducted with 120 estimations of each configuration towards random screws on the battery pack. Moreover, half of them are randomly selected to set up obstacles (disassembled screws) next to the target screws for verifying the validity of neural predicates.

The errors of position and pose estimation and the success rate of subsequent insertions are shown in Table 2. Position calculates the coordinates of the screw in the world coordinate. Orientation calculates the angles of the normal vectors of fitting planes in the world coordinate. In the 'YOLO only' configuration, the system detects the screw position in the RGB image through YOLOv5. Then the system finds 3 points on the screw surface, and calculates the normal vector of the fitting plane directly without RANSAC algorithm. In the 'YOLO w/ KF' configuration, the system introduces the Kalman Filter based on 'YOLO only' and obtains more accurate positions through multiple observations. In the 'YOLO w/ RANSAC' configuration, the system introduces proposed RANSAC-based pose estimation to calculate the orientation of the screw based on YOLOv5 detection. In YOLO w/ KF & RANSAC configuration, the system uses the proposed Kalman Filter and RANSAC algorithm simultaneously based on YOLOv5 detection.

It can be found that after the introduction of the RANS-AC-based pose estimation, the position and pose estimation error is significantly reduced, and the success rate of subsequent insertion also improves a lot, indicating that more accurate orientation estimation contributes to the implementation of insertion. After Kalman filtering is introduced, the accuracy of position and pose estimation is further improved, and finally the insertion success rate reaches 100%. In sum, our proposed screw position and pose estimation based on visual perception can well meet the strict requirements of intelligent disassembly of EVBs.

Table 2
Accuracy and average error of vision-based position and pose estimation.

	Success rate (%)	Average error			
		Position (mm)			Orientation (rad)
		x	y	z	
YOLO w/ KF & RANSAC	100.0	0.38	0.37	0.55	0.017
YOLO w/ RANSAC	98.3	0.48	0.42	0.56	0.028
YOLO w/ KF	25.8	0.48	0.59	0.58	0.372
YOLO only	12.1	0.55	0.61	0.61	0.428

As for the experiments of neural predicates, relatively good results are obtained. For the neural predicate ***target_clear()***, the results show that the system can correctly distinguish the situation with or without obstacles every time and execute the **Push** primitive to clear the obstacles as appropriate. For the neural predicates ***target_aim()***, the results show that the system can trigger the **Mate** primitive to obtain more accurate position and pose estimations if needed.

5.2. Insertion based on force perception

In the above experiments, the validity of our proposed screw position and pose estimation based on visual perception is fully verified. However, in order to meet the stringent requirements of robustness in the industry, our system also includes an MDP decision model based on force perception. To evaluate the effectiveness and robustness of the MDP model as well as the accuracy of active exploration proposed in this paper, random positions within 2.5–9.5 mm from the real position of the screw serve as the estimated screw position (to prevent being directly successfully clamped or too far beyond the achieved positioning accuracy to be meaningful), and 40 insertion experiments were performed using this method.

The force feedback of **Fumble** primitive and **Search** primitive starting from a partially/not clamped state are shown in Fig. 13 respectively. The color bands in the middle of Fig. 13 indicate different states. As for the **Fumble** primitive which is used to clamp the screw in partially clamped state, due to the compressed spring in the disassembly actuator, the sleeve and the target screw remain in close contact without abrupt force changes in horizontal and vertical directions. At the moment the sleeve clamps the target screw, the compressed spring is released and a force change in vertical direction can be observed. Correspondingly, a greater force change in the horizontal direction can be observed after successfully clamping. As for the **Search** primitive which is used to clamp the screw in not clamped state, the contact between the sleeve and the outer margin of the target screw, during the **Search** primitive execution process, results in an abrupt force change in horizontal direction. Subsequently, **Re-insert** primitive is executed to clamp the target screw. In sum, the **Fumble** primitive and the **Search** primitive are executed based on different force perception of clamping process.

Theoretically, the decision model based on force feedback should be quite accurate due to the fact that only a few steps are required to realize the clamping. But in practice the judgment results of the active exploration of the contact state are not completely reliable. To assess the impact of such misjudgment, the confusion matrix for active exploration of contact states is obtained in Table 3. From Table 3, it can be found that the overall accuracy of the judgment is relatively high, and misjudgments that are prone to occur include judging ***success*** as ***part*** or judging ***part*** as ***lost***. Fortunately, if the state ***success*** is judged to be ***part***, the **Fumble** primitive will be executed. As long as a force mutation is detected, the active search will be performed again. Similarly, if the state ***part*** is judged to be ***lost***, the **Search** primitive will be executed, and it generally will be transformed to the state ***lost***.

Table 3
Confusion matrix for active exploration.

Contact state		Predicted		
		success	part	lost
Actual	success	40	13	2
	part	0	31	16
	lost	0	1	41

Therefore, successful clamping was finally achieved in all 40 experiments carried out. In most cases, insertion can be completed simply by executing the **Fumble** primitive. In the remaining cases, basically only one **Re-insert** operation is required to achieve successful clamping. In the few cases where multiple **Re-insert** operations are required, it is the inaccurate screw positioning based on force perception that causes redundant operations. Further improvements need to be made in the future.

5.3. Efficiency

In order to verify the efficiency of the proposed methods, an efficiency experiment is conducted to test the average running time (ART) of each link. The efficiency experiment is divided into two groups, with and without obstacles, each of which is tested 50 times repeatedly. The test links are divided into four parts: autonomous planning based on NeuroSymbolic, vision-based algorithms, force-based algorithms and

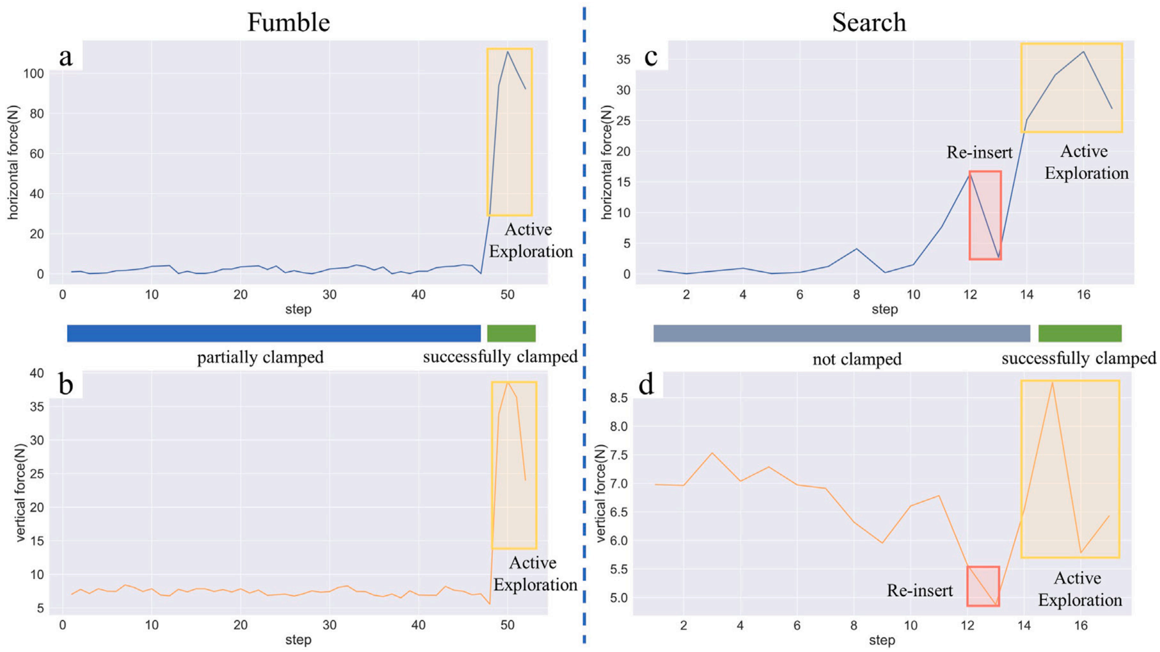

Fig. 13. (a)(b) force feedback during the execution of the **Fumble** primitive; (c)(d) force feedback during the execution of the **Search** primitive(blue: partially clamped, gray: not clamped, green: successfully clamped).

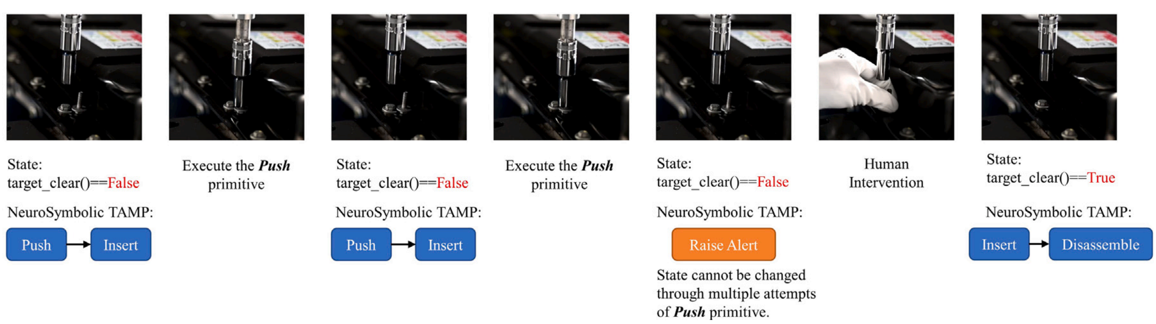

Fig. 14. Experiment for human–robot collaboration.

the whole process. The NeuroSymbolic TAMP includes the process of generating primitive sequences and the judgment of neural predicates. Moreover, vision-based algorithms include the process of vision-based position and pose estimation and Kalman filtering process, while force-based algorithms include the process of robust insertion algorithms based on force perception. The whole process indicates the total process of disassembling screws. The experimental results are shown in the Table 4.

Table 4
Average running time of each link.

Test link	ART(s) (w/ obstacles)	ART(s) (w/o obstacles)
NeuroSymbolic TAMP	0.56	0.53
Vision-based Algorithms	2.36	2.47
Force-based Algorithms	3.81	3.55
Whole Process	11.23	7.64

It should be noted that the most insertion contact states during the experiment are successfully clamped, while the rest are partially clamped. Cases with obstacles increase the running time due to the **Push** primitive compared to cases without obstacles. Many technologies can be optimized to improve efficiency in the future.

5.4. Human–robot collaboration

We experiment with human–robot collaboration, as shown in Fig. 14. To conduct experiments for human–robot collaboration, we simplified the action of **Push** primitive so that the robot cannot successfully clear the artificially designed obstacle. Through experiments, we verified that the system has a high level of interpretability. The NeuroSymbolic TAMP ensures that the operations of the robot are executed within the logic framework. Moreover, when the system finds that it cannot independently change the current state by performing actions, it can request manual help by itself, which ensures safe operations in human–robot scenarios. For example, as shown in Fig. 14, when the robot executes the **Push** primitive multiple times but fails to successfully clear the obstacle, the robot will request human assistance to complete the obstacle clearing and continue to work. In this way, workers can know what the robot is doing and what its purpose is.

6. Conclusion and future work

Aiming at the intelligent disassembly of EVBs in an unstructured disassembling environment, an intelligent human–robot disassembling process and a framework of knowledge-driven flexible human–robot hybrid disassembly line are proposed for efficient, flexible and green disassembly of EVBs. To further elaborate the key technologies of the disassembly line, this paper focuses on the implementation of high-accuracy screw disassembly and an autonomous disassembly planning system based on NeuroSymbolic. The NeuroSymbolic TAMP system ensures the advantages of autonomy, interpretability, learnability and scalability for the operations of the robot. The implementation of high-accuracy screw disassembly is realized by designing a new disassembly actuator and adopting vision-based position and pose estimation and robust force-based screw insertion algorithms. Experiments show that our proposed methods are reliable and efficient in the context of human–robot collaboration, which provides technical support for the intelligent disassembly of EVBs.

In the future, there are several aspects to be improved or further investigated.

1. In the future, more disassembly primitives and predicates need to be designed to deal with more complex disassembly tasks of EVBs, such as lifting and transferring battery modules. Furthermore, the continual learning of neural predicates is another focus of the following research, which can constantly improve the accuracy of neural predicates and adapt to the new changes in the environment.
2. Due to the limitation of data acquisition conditions in the object detection algorithm of fasteners, the relevant data sets need to be expanded and optimized to further improve the detection performance and generalization.
3. Robust force-based screw insertion algorithms need to be optimized to further improve the judgment of the contact state. Furthermore, force-based models and algorithms can be extended to other disassembling tasks.
4. After verifying the feasibility of the system in the laboratory, the balancing problem of manual and automatic workstations in the disassembly line needs to be further investigated, which requires comparing the efficiency of manual and automatic workstations and making comprehensive decisions. It is worth mentioning that such modules have been reserved in our system design to support expansion and continuous adjustment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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